Hinge-Moment Estimation for Fighter Aircraft Flight Control Surfaces: A Comparative Semi-Empirical and Panel-Method Framework with a Preliminary Supersonic Extension

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Abstract

Early evaluation of the on-board Flight Control System (FCS) is increasingly recognized as a critical research topic for estimating aircraft mass distribution and power requirements during the conceptual design phase, particularly for fighter aircraft and in the context of the ongoing transition toward more-electric actuation architectures. Estimating FCS power requirements depends on hinge-moment predictions, traditionally obtained through semi-empirical correlations that assume linear aerodynamics based on years of historical data and lack reliability under critical conditions and for modern three-dimensional control-surface geometries. At the same time, high-fidelity CFD remains too computationally expensive for iterative conceptual design. This paper presents a parametric, CPACS-based framework that estimates aileron hinge moments using two complementary methods: an automated implementation of Roskam's semi-empirical methodology and a Tornado vortex-lattice panel approach. Both methods are validated against NASA flight-test data and a reference simulation for the F/A-18 Systems Research Aircraft across subsonic and supersonic flight conditions. In the subsonic regime, both methods achieve root-mean-square errors within approximately 10-20% of the reference simulation, while both break down beyond Mach 1 due to the invalidity of the Prandtl-Glauert compressibility correction. Building on this outcome, the paper additionally reports a preliminary extension of the vortex-lattice formulation to supersonic flow using Miranda's generalized supersonic kernel, intended to extend the framework's validity range beyond

the transonic barrier. The proposed framework is intended to support early-stage FCS sizing decisions at a computational cost compatible with the iterative evaluation cycles of aircraft conceptual design.

Keywords: hinge moment, flight control system, vortex lattice method, aircraft conceptual design, CPACS, supersonic aerodynamics

1 Introduction

During aircraft conceptual design, early evaluation of on-board systems' configuration has been identified as a promising research topic, since early definition of power management and architecture leads to cascading effects on the overall design [1]. This is particularly relevant for fighter aircraft, whose aggressive maneuvering imposes demanding loads on the Flight Control System (FCS), and for the ongoing transition toward more-electric architectures, which further increases the need for accurate early-stage load and power estimation [2].

Estimating the overall power required by the FCS is challenging: it depends on actuator loads derived from hinge-moment predictions, which typically rely on semi-empirical correlations involving flight-envelope functions, control-surface geometry, and deflections [3, 4]. Such correlations are fast but assume linear, attached-flow aerodynamics and are poorly suited to the complex three-dimensional geometries of modern fighters; in contrast, high-fidelity computational fluid dynamics (CFD) remains too computationally expensive for the iterative sweeps required at the conceptual stage [5].

This paper presents a parametric framework that compares a semi-empirical method (Roskam) and a panel method (Tornado vortex-lattice method [6]) within a common geometry description based on the Common Parametric Aircraft Configuration Schema (CPACS) [7], and reports a comparative validation against NASA flight-test data for the F/A-18 Systems Research Aircraft (SRA). Beyond the validated subsonic-to-supersonic comparison, this paper extends the original work [8] with a dedicated section on an ongoing development: a linearized supersonic correction intended to recover method validity beyond Mach 1.

2 Background

For a control surface, the aerodynamic moment about the hinge line is commonly modeled, within the linear regime, as

$$M_h = q_\infty \, S_f \, c_f \, (C_{h_0} + C_{h_\alpha} \, \alpha + C_{h_\delta} \, \delta) \, , \tag{1}$$

where q_∞ is the free-stream dynamic pressure, S_f and c_f are the control-surface reference area and mean chord, α is the angle of attack, δ is the control-surface deflection, and C_{h_α}, C_{h_δ} are the hinge-moment derivatives relative to the same angles. The hinge-moment coefficient component C_{h_0} is either negligible for a symmetric section

or strongly dependent on airfoil shape; therefore, reliably estimating C_{h_α} and C_{h_δ} is the central problem this work addresses.

Roskam's semi-empirical method [3] obtains C_{h_α} and C_{h_δ} through a chain of corrections (Mach number, aerodynamic balance, thickness, three-dimensional effects) extracted from digitized experimental charts. The Tornado vortex-lattice method [6], on the other hand, is an inviscid panel method that computes the hinge moment directly by integrating panel-wise aerodynamic forces about the hinge axis. Both methods assume linear, attached-flow aerodynamics and apply, by default, a Prandtl-Glauert compressibility correction that is only valid for $M < 1$.

3 Methodology

The overall methodology adopted in this work is illustrated in Fig. 1.

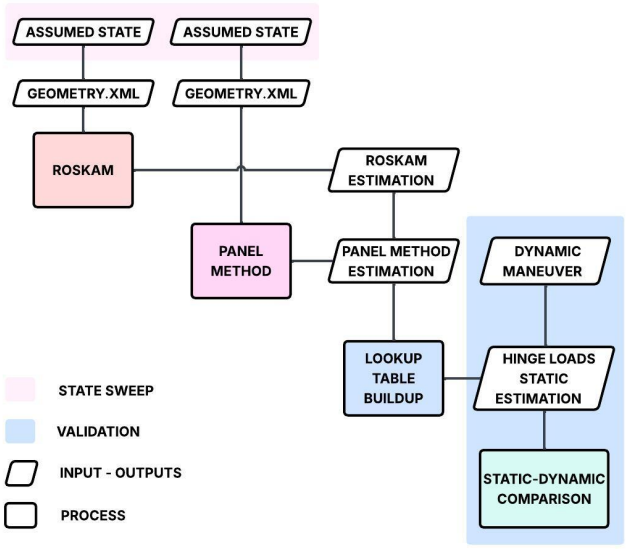


Fig. 1 Modular methodology adopted in the original work [8], in an XSDM format, with inputs coming from the top and bottom sides of each square (or process) and outputs on the sides, highlighting the flight condition (state) sweep as well as the final comparison from a dynamic maneuver

3.1 Roskam Semi-Empirical Implementation

An automated MATLAB implementation of Roskam's method (Volume VI [3]) was developed to remove the need for manual chart reading. The aforementioned derivatives are obtained as

$$C_{h_\alpha} = \left[\frac{AR\cos\Lambda_{c/4}}{AR + 2\cos\Lambda_{c/4}}\right] C_{h_{\alpha,M}} + \Delta C_{h_\alpha}, \tag{2}$$

$$C_{h_\delta} = \cos\Lambda_{c/4}\cos\Lambda_{hl}\left[C_{h_\delta,M} + \alpha_\delta C_{h_\alpha,M}\frac{2\cos\Lambda_{c/4}}{AR + 2\cos\Lambda_{c/4}} + \Delta C_{h_\delta}\right], \tag{3}$$

where $C_{h_\alpha,M}$ and $C_{h_\delta,M}$ include the Mach-corrected, balance-corrected two-dimensional derivatives, and ΔC_{h_α}, ΔC_{h_δ} account for finite-span effects. Relevant Roskam charts were digitized in MATLAB (see [8] for the full correction chain and geometrical variables definition).

3.2 Tornado Panel-Method Implementation

The open-source Tornado MATLAB tool [6] was adapted to share the same geometric input structure as the Roskam module. For each control-surface panel, the force vector $\vec{F}_i$ is extracted from the vortex-lattice solution and the moment contribution about the hinge axis $\hat{h}$ is computed and summed over all flap panels,

$$M_h = \sum_{i=1}^{n}\left(\vec{r}_i \times \vec{F}_i\right)\cdot\hat{h}, \qquad C_h = \frac{M_h}{q_\infty S_f c_f}. \tag{4}$$

3.3 CPACS-Based Geometry Interface

Geometry is defined in a standardized CPACS [7] file and automatically used to assemble a shared `Geo` structure consumed by both estimation modules, eliminating manual re-entry of parameters across tools. A limited set of Tornado-specific solver settings (panel counts, mesh type) without a native CPACS equivalent is stored in a dedicated `toolspecific` branch.

4 Results

4.1 Test Case Definition

The reference aircraft is the F/A-18 SRA, for which NASA flight-test data [9] and a more high-fidelity simulation of [10] are available. Table 1 summarizes the wing and aileron geometry used as input to both methods.

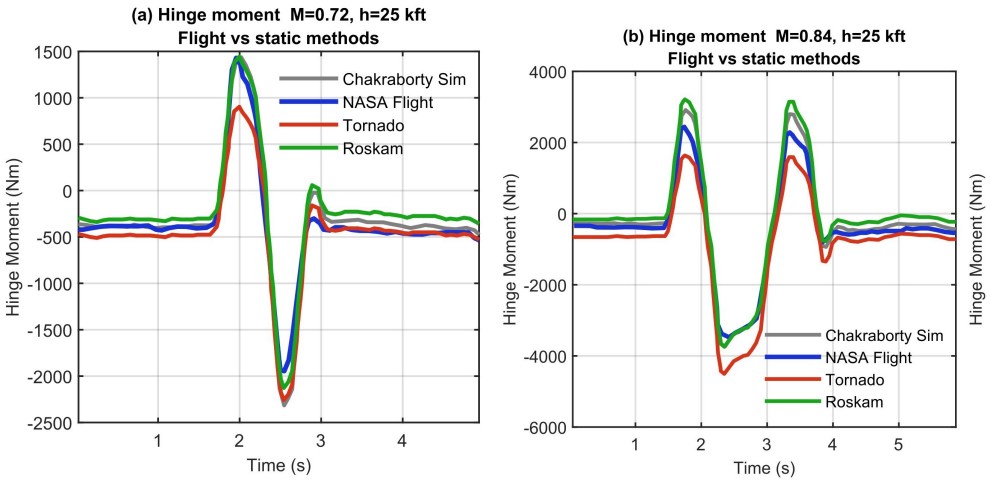
Table 1 F/A-18 SRA geometry inputs used for hinge-moment estimation [9, 11]

Parameter	Description	Value
b	Wing span	12.31 m
S	Wing reference area	37.16 m^2
S_f	Aileron surface area	1.135 m^2
$\Lambda_{c/4}$	Quarter-chord sweep	20 deg

Four flight conditions were selected to match previous studies [9, 10]: two subsonic maneuvers (Mach 0.72 and 0.84, 25 kft) within the linear validity range of both methods, and two supersonic maneuvers (Mach 1.58 and 1.60, 34 kft) used to characterize method breakdown and investigate possible supersonic corrections implementation.

4.2 Subsonic Flight Conditions

The results obtained across the subsonic conditions obtained with the methodologies discussed are shown in Fig. 2 and summarized in Table 2, considering the root-mean-square (RMS) and mean-absolute error (MAE) of both methods against NASA flight data, together with the reference simulation.



(a) (b)

Fig. 2 Hinge-moment estimation comparison with NASA flight data, reference simulation, Tornado, and Roskam estimates: (a) Mach 0.72 roll doublet maneuver; (b) Mach 0.84 aileron reversal maneuver

Table 2 RMS and MAE summary for the two subsonic cases. All values in Nm

	Tornado		Roskam		Reference [10]
Case	RMS	MAE	RMS	MAE	RMS
Mach 0.72	199.4	142.4	184.3	157.3	178.7
Mach 0.84	524.1	450.1	459.5	383.0	280.0

Both traditional and panel methods track the flight data closely at Mach 0.72, with errors within approximately 12% of the high-fidelity reference. At Mach 0.84 the discrepancy grows for all methods, consistent with increasing transonic compressibility effects that are only partially captured by the Prandtl-Glauert correction.

4.3 Supersonic Flight Conditions

Beyond Mach 1, the Prandtl-Glauert correction used by default in both methods becomes physically undefined. Therefore Tornado's incompressible formulation collapses to a near-zero hinge moment regardless of the aileron input, while Roskam retains only a small, quasi-steady offset; neither reproduces the ±6000 Nm oscillatory peaks observed in the flight data (RMS errors of approximately 3300-3600 Nm, Table 3).

Table 3 RMS and MAE summary for the two supersonic cases. All values in Nm

	Tornado		Roskam	
Case	RMS	MAE	RMS	MAE
Mach 1.58	3322.1	2554.9	3600.8	2739.4
Mach 1.60	3339.2	2572.5	3635.9	2804.1

5 Supersonic Extension: Miranda Kernel Implementation

The results of Section 4 show that the Prandtl-Glauert-based formulations used by the baseline implementations cannot be extended directly beyond Mach 1. To investigate whether the vortex-lattice framework can instead be extended to supersonic flow, Tornado was modified using the generalized swept-horseshoe formulation developed by Miranda for subsonic and supersonic applications [12, 13].

The formulation is derived from linearized, inviscid, compressible potential flow. In the supersonic regime, the governing small-disturbance equation changes from elliptic to hyperbolic and can be written as

$$\beta^2 \frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \phi}{\partial y^2} - \frac{\partial^2 \phi}{\partial z^2} = 0, \qquad \beta = \sqrt{M_\infty^2 - 1}. \tag{5}$$

A key consequence of the hyperbolic character of the equation is the finite domain of dependence associated with the Mach cone. Two implementations were therefore investigated for the present control-surface application.

1. **Mach Cone Masking (`SUP CONE MASK`):** Panel influence is restricted according to the supersonic domain of dependence of each control point. Only source panels located within the corresponding upstream Mach cone are retained in the aerodynamic influence calculation [12, 13]. In the present implementation, this condition is expressed as
$$X_k^2 - \beta^2 (Y_k^2 + \bar{z}^2) > 0. \tag{6}$$
This geometric treatment reproduces the finite supersonic domain of dependence, but it substantially underpredicts the measured hinge-moment response, with peak

magnitudes of approximately $2,500$ Nm compared with approximately $6,000$ Nm in the flight data.

2. **Miranda Kernel with Self-Residual (`SUP MIRANDA KJ`):** The second implementation replaces the purely geometric influence truncation with the generalized supersonic horseshoe-vortex formulation of Miranda. In addition to the supersonic induced-velocity kernel, the formulation introduces a diagonal self-residual term associated with the generalized principal part of the vortex-induced velocity. For subsonically swept bound vortex legs ($\beta^2 > t^2$), the implemented residual is

$$w^* = -\frac{1}{2}\frac{\sqrt{\beta^2 - t^2}}{\Delta x_{\text{patch}}}, \qquad \Delta x_{\text{patch}} = \frac{\text{Panel Area}}{2s}, \tag{7}$$

where s is the transverse semi-span and t is the local sweep parameter. This residual term is a fundamental component of Miranda's generalized vortex-lattice formulation rather than an empirical correction [12, 13].

The two formulations were evaluated against the NASA flight-test data for the F/A-18 SRA at $M = 1.58$ and $M = 1.60$. As shown in Fig. 3, the cone-masking approach captures only about 40% of the measured dynamic peak magnitude. In contrast, the `SUP MIRANDA KJ` formulation reproduces the observed oscillatory response, including its phase and characteristic dynamic frequency, and recovers the correct order of magnitude of the hinge moment. The predicted loads, however, remain higher than the flight-test values at the largest aileron deflections.

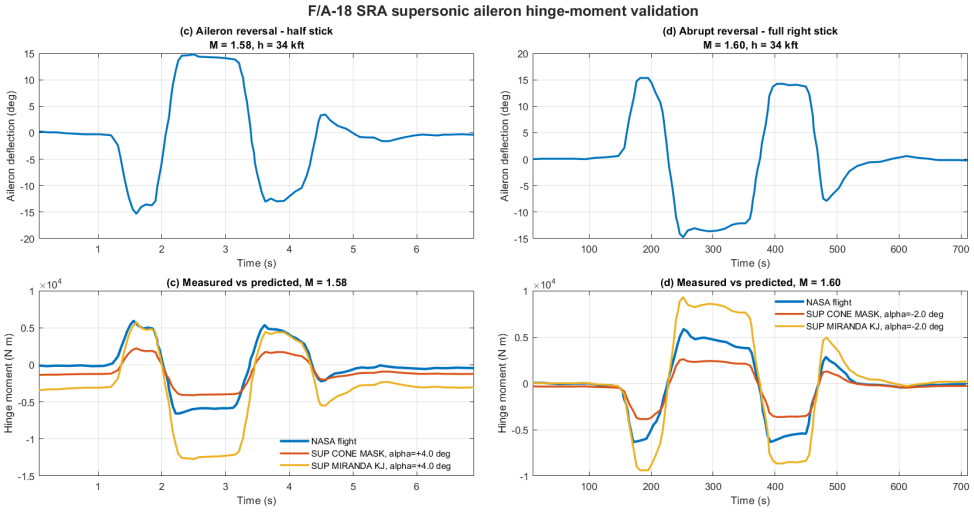


Fig. 3 Supersonic aileron hinge-moment validation for $M = 1.58$ (half stick) and $M = 1.60$ (full right stick) maneuvers. NASA flight data are compared with the Mach-cone masking and Miranda generalized supersonic-kernel implementations

5.1 Physical Interpretation of Peak Discrepancies

The Miranda formulation improves the supersonic prediction relative to the baseline methods, although a significant discrepancy remains at large control-surface deflections. At $M = 1.58$ and $|\delta| \approx 15°$, the predicted hinge moment reaches approximately $-12,800$ Nm, compared with a measured peak of approximately $-6,500$ Nm.

This overprediction can be attributed to both the assumptions of the generalized vortex-lattice formulation and its numerical sensitivity. Miranda's method is based on linearized, inviscid potential-flow theory and therefore does not explicitly capture nonlinear shock and expansion-wave interactions, viscous effects, boundary-layer development, or flow separation [12, 13]. At large deflections, these effects may produce a nonlinear pressure response and reduce the aerodynamic loads relative to the model prediction. In addition, the computed surface loading is sensitive to lattice density and vortex distribution, which is particularly relevant for hinge moments that depend on the local pressure distribution over the control surface [13].

6 Discussion

The two baseline approaches provide comparable hinge-moment estimates under the investigated subsonic conditions. At $M = 0.72$, both reproduce the measured dynamic response with RMS errors close to those of the higher-fidelity reference simulation, while the error increases at $M = 0.84$, reflecting the growing influence of compressibility effects as the transonic regime is approached. Beyond Mach 1, the limitations of the baseline formulations become structural rather than merely quantitative. The Prandtl-Glauert correction is not applicable in the supersonic regime, motivating the replacement of the subsonic influence formulation rather than further calibration of the existing correction factor.

The Miranda-based extension demonstrates that a generalized supersonic vortex-lattice formulation can be integrated into the existing Tornado framework and applied to the same CPACS-derived geometry. Mach-cone masking captures the finite supersonic domain of dependence, but its strong underprediction shows that geometric truncation alone is insufficient. Including the generalized Miranda kernel and self-residual term instead restores a substantially more realistic dynamic response.

The remaining overprediction at large aileron deflections reflects both the limitations of the linearized inviscid formulation and the numerical sensitivity of the lattice discretization. Nonlinear and viscous effects are not explicitly represented, while lattice density and vortex distribution can affect the predicted surface loading [13].

Finally, the validation is limited to a single aircraft configuration and four flight conditions. The dynamic reconstruction also uses the best-fit angle-of-attack selection described in [8], introducing some circularity. The reported subsonic errors should therefore be interpreted as configuration-specific performance indicators rather than fully blind validation metrics.

7 Conclusions and Future Work

A CPACS-based framework for aileron hinge-moment estimation was presented by integrating an automated Roskam semi-empirical implementation and a Tornado vortex-lattice formulation within a common geometry interface. Validation against F/A-18 SRA flight-test data showed comparable performance under the investigated subsonic conditions, with increasing discrepancies as the transonic regime was approached.

Beyond Mach 1, the Prandtl-Glauert-based formulations proved unsuitable. The Miranda-based extension recovered the correct dynamic response and order of magnitude, though overprediction at large deflections indicates a preliminary rather than fully validated implementation, warranting a dedicated mesh-convergence study.

Future work should focus on mesh and Mach-cone sensitivity studies and broader validation across supersonic Mach numbers and control-surface deflections. Higher-fidelity viscous simulations could then help distinguish discrepancies due to discretization from nonlinear and viscous phenomena. Longer-term developments include full CPACS compliance, coupling with a six-degree-of-freedom flight-mechanics model, and integration with actuator power models for early electrical-system sizing in more-electric fighter architectures.

Abbreviations. The following abbreviations are used in this manuscript:

CFD	Computational Fluid Dynamics
FCS	Flight Control System
CPACS	Common Parametric Aircraft Configuration Schema
SRA	Systems Research Aircraft
RMS	Root Mean Square
MAE	Mean Absolute Error

Acknowledgements. The authors acknowledge Linköping University, Department of Management and Engineering, for supporting this work.

Statements and Declarations

Funding. This research received no external funding.

Competing interests. The authors declare no conflicts of interest.

Ethics approval and consent to participate. Not applicable.

Data availability. The data and code supporting the reported results are available from the corresponding author upon reasonable request.

Author contributions. Conceptualization, Andrea Vitale and Erick Espinosa Juárez; methodology, Andrea Vitale; software, Andrea Vitale and Leone Tuci; validation, Andrea Vitale; formal analysis, Andrea Vitale; investigation, Andrea Vitale and Erick Espinosa Juárez; writing—original draft preparation, Andrea Vitale; writing—review and editing, Andrea Vitale and Erick Espinosa Juárez; supervision, Erick Espinosa Juárez. All authors have read and agreed to the published version of the manuscript.

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